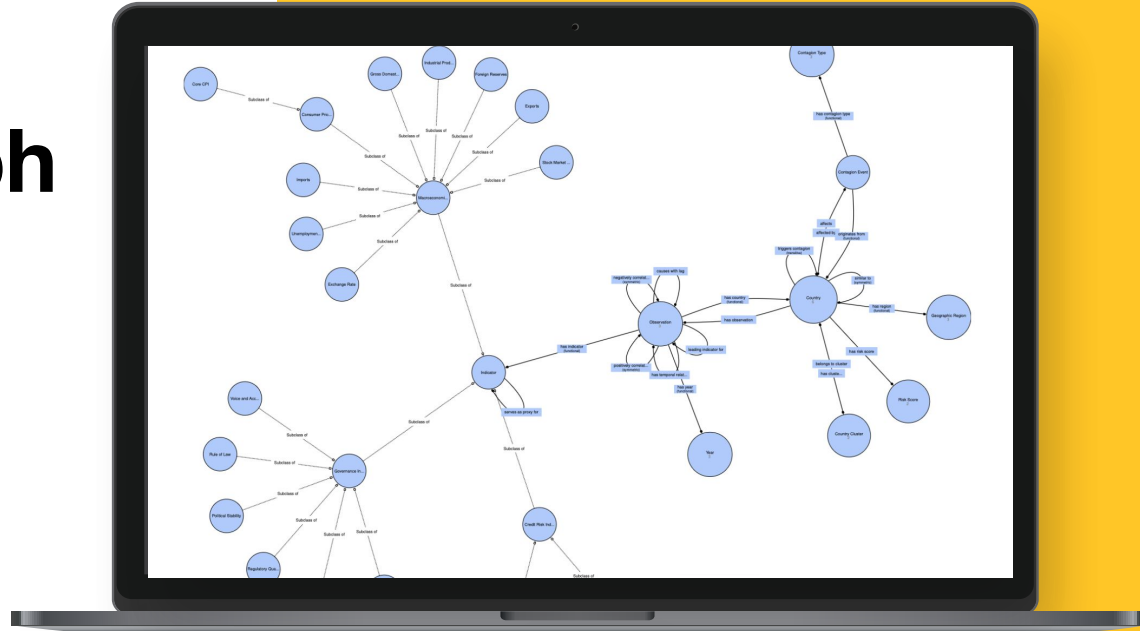


GEMR:KG Global Emerging Market Risk Knowledge Graph

Adishesh Gonibeed Ravishankar,
Hemakshi Pandey,
Hitesh Kolluru,
Lalitha Shreya Vanam,
Likitha Sathvik Potineni, and
Dr. Srividya Bansal



Agenda

01. Emerging Markets and current analytical practices

02. Problem

03. Solution

04. Technical Framework

05. Results

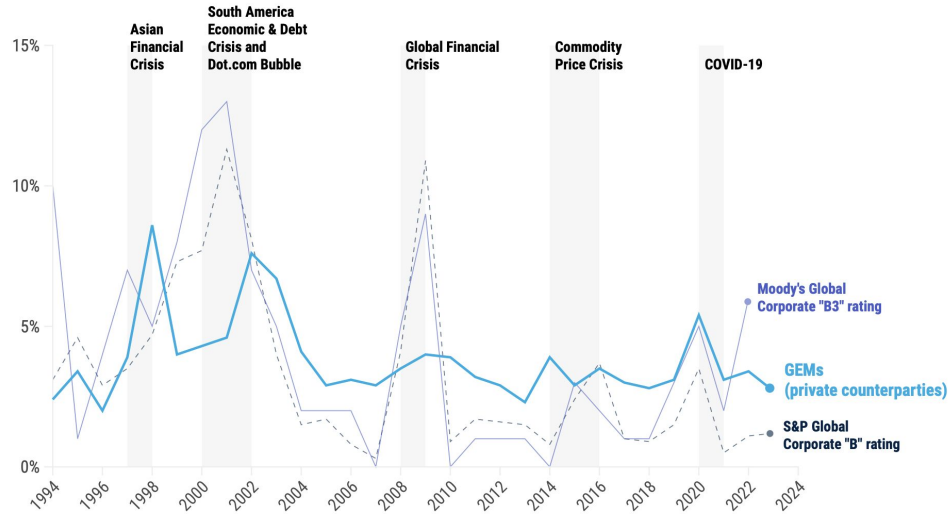
06. Challenges and Future Research

Emerging Markets

- ➔ **Fast Growing Economies**
- ➔ **Integrate into Global Markets**
- ➔ **Higher Volatility , Political Risks**
- ➔ **Less mature financial systems**
- ➔ **High Growth Potential ~ More Risk**



FIGURE 1
Annual Default Rates, GEMs Versus Comparators
In percent



Lower Volatility during Crisis?

How can Emerging markets have lower peak default rates during major global crises compared to their similarly-rated corporate counterparts

Problem

MSCI 

MSCI Emerging Markets Index

Bloomberg

**Emerging
Markets**

J.P.Morgan
Asset Management



Emerging
Markets
Institute

北京師範大學
新興市場研究院

No unified and STANDARD Index
indicating risk

Current indicators vary marginally

Too many factors and macroeconomic
indicators not covered

Inaccessible & Siloed Data

Difficulty in extracting real-time
actionable insights.

Solution

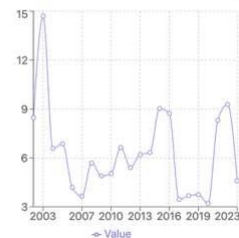
KG



Brazil

Select Indicator:

CPI Price Pct yearoveryear seasonally adjusted



SPARQL Query

Run Query

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX gnr: <http://sem.wpi.org/ontology#>

SELECT ?countryName ?yearInt ?stockValue ?targetYearInt ?avgDefaultRate
WHERE {
  # 1. Source: Stock Market (Year T)
  ?obsStock a gnr:StockMarket_Index_LCW ;
    gnr:hasCountry ?c ;
    gnr:hasYear ?yearEntity ;
    gnr:observationValue ?stockValue .

  # 2. Extract Integer Year
  ?yearEntity gnr:yearValue ?yearLiteral .
  BIND(xsd:integer(STR(?yearLiteral)) AS ?yearInt)

  # 3. Calculate Target (T+1)
  BIND(?yearInt + 1 AS ?targetYearInt)

  # 4. Target: Private Default Rate (Year T+1)
  ?targetYearEntity gnr:yearValue ?
  ?targetYearLiteral .
  FILTER(xsd:integer(STR(?targetYearLiteral)) = ?
  ?targetYearInt)

  ?obsDefault a gnr:PrivateDefaultRate ;
    gnr:hasCountry ?c ;
    gnr:hasYear ?targetYearEntity ;
    gnr:observationValue ?defaultRate .
```

Use Cases

Temporal Analysis

1. Early Warning (Stock -> Default)

Stock Market (t) vs Default Risk (t+1)

2. Political Stability -> GDP

Impact of Political Stability (t) on GDP (t+1)

3. Default -> Recovery

Stock Market (t) vs Real Economy GDP (t+1)

4. GDP Growth Tracker

Calculated annual GDP growth (t vs t-1)

Results

?countryName	?yearInt	?stockValue	?targetYearInt	?avgDefaultRate
Brazil	2002	16.75262	2003	5.7971015
Brazil	2003	21.02394	2004	0.0
Brazil	2004	32.54595	2005	1.6949153
Brazil	2005	40.17986	2006	0.0
Brazil	2006	55.5775	2007	1.9607843
Brazil	2007	77.63148	2008	1.754386
Brazil	2008	80.6549	2009	6.1538463
Brazil	2009	76.97017	2010	0.0
Brazil	2010	98.10249	2011	0.0
Brazil	2011	89.41041	2012	3.6363637
Brazil	2012	86.93919	2013	4.255319
Brazil	2013	78.32967	2014	6.976744
Brazil	2014	76.7433	2015	1.7241379
Brazil	2015	72.58238	2016	7.9365077
Brazil	2016	77.43924	2017	3.5714285
Brazil	2017	99.31608	2018	0.0
Brazil	2018	119.2345	2019	0.0
Brazil	2019	146.8797	2020	3.2258065
Brazil	2020	144.0616	2021	1.4492754
Brazil	2021	170.4105	2022	0.0
Brazil	2022	159.9459	2023	1.2048193
China	2006	50.67089	2007	6.25
China	2007	133.7714	2008	0.0
China	2008	96.68721	2009	3.7735846

Capture macroeconomic complexities via a unified semantic model.

Handle Heterogeneity

Complex Analysis

Tool for Economists and Analysts

Datasets



WORLD BANK GROUP

Global Economic Monitor (GEM):

A daily monitor of global economic developments and high-frequency data.

Indicators:

- GDP and Growth Rates
- Inflation (CPI)
- Exchange Rates
- Trade Balances

Global Emerging Markets Risk Database (GEMS):

A dataset of defaults and loss statistics for loans in emerging markets.

Indicators:

- Default Rates
- Recovery Rates
- Private vs Public
- Sector Analysis

Worldwide Governance Indicators (WGI):

Aggregate indicators for over 200 countries across six key dimensions of governance

Indicators:

- Political Stability
- Control of Corruption
- Rule of Law
- Regulatory Quality

Sources :

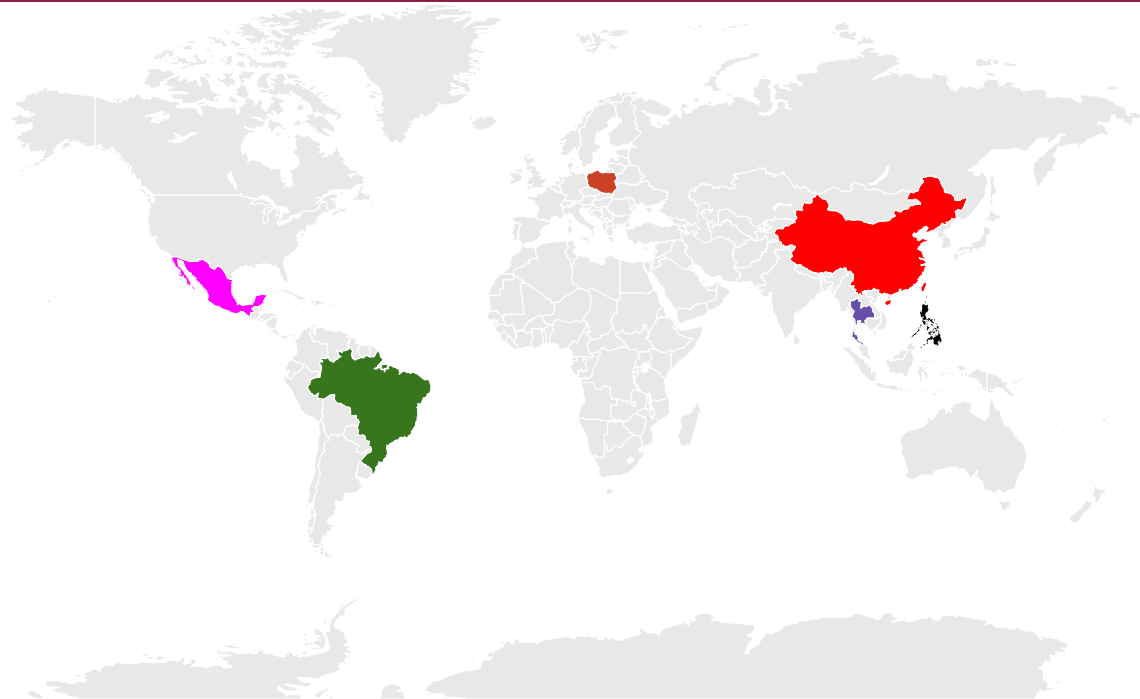
Global Economic Monitor — Data Catalog, "Global Economic Monitor— Data Catalog. Available: <https://datacatalog.worldbank.org/search/dataset/0037798/Global-Economic-Monitor>

World Bank Group, "Global Emerging Markets Risk Database (GEMS) Consortium," Mar. 2025. Available: <https://www.worldbank.org/en/about/unit/brief/gems>

"Worldwide Governance Indicators," The World Bank. Available: <https://www.worldbank.org/en/publication/worldwide-governance-indicators>

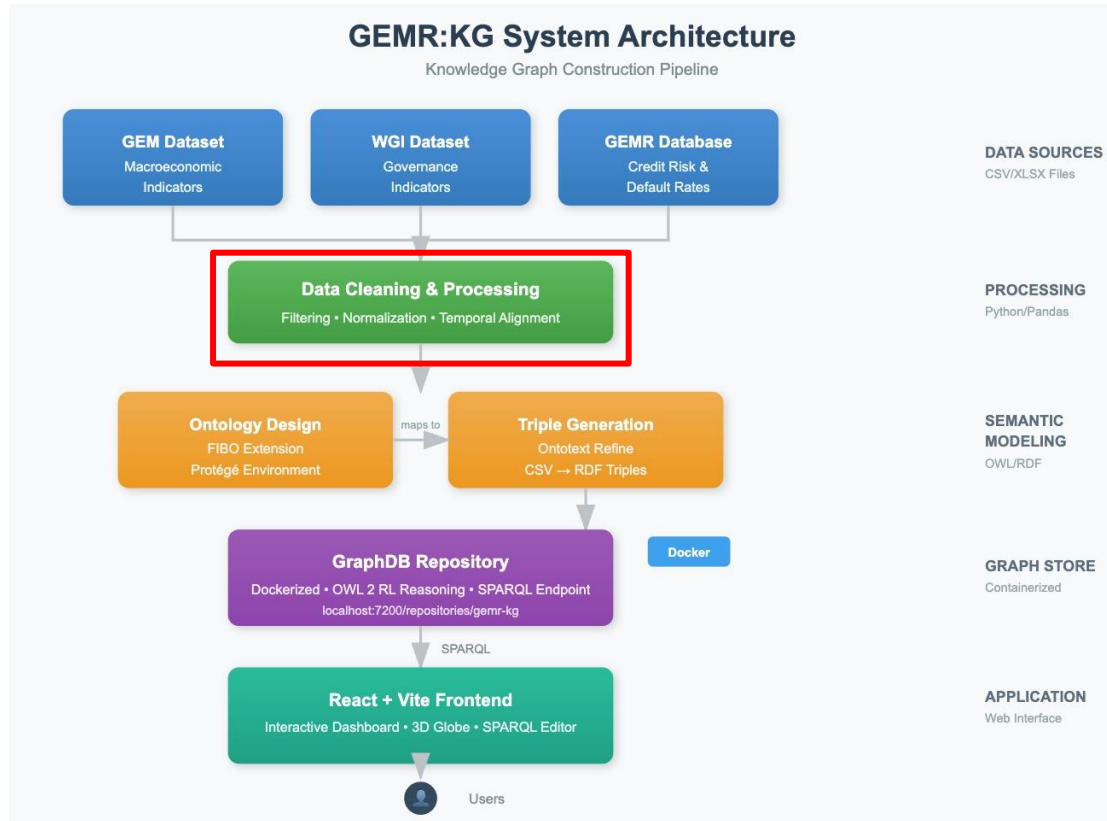
Countries Covered: 6

Timeframe : 2002 - 2023

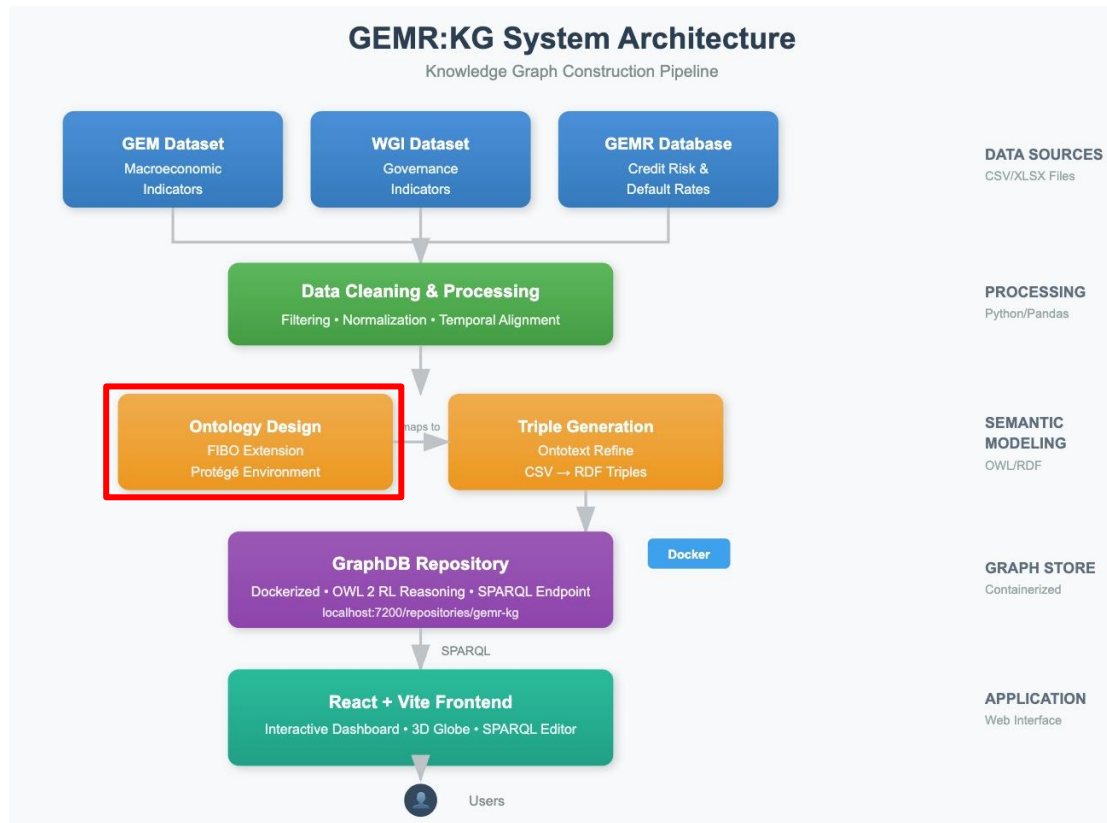


- Brazil 
- China 
- Mexico 
- Philippines 
- Poland 
- Thailand 

Technical Framework



Technical Framework



Ontology Design

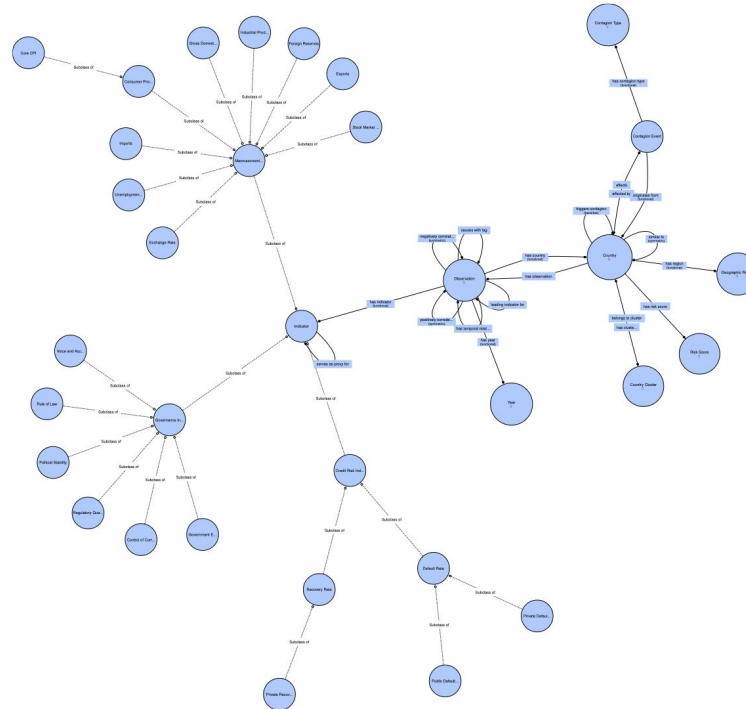


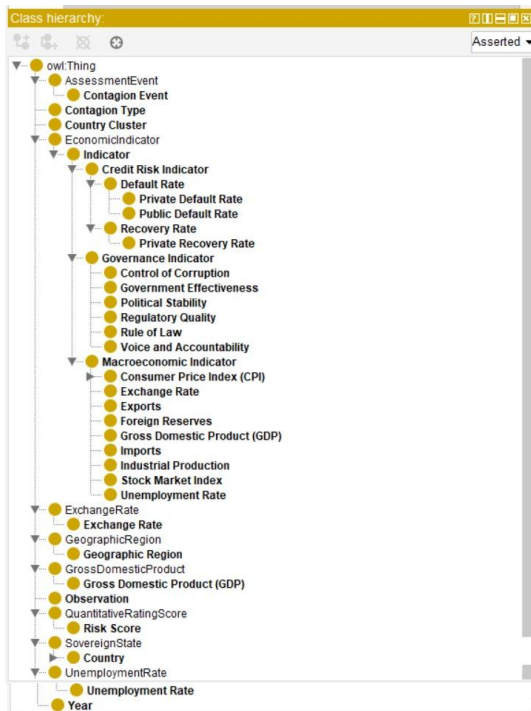
Figure 1. GEMR-KG Ontology Structure

Ontology Design

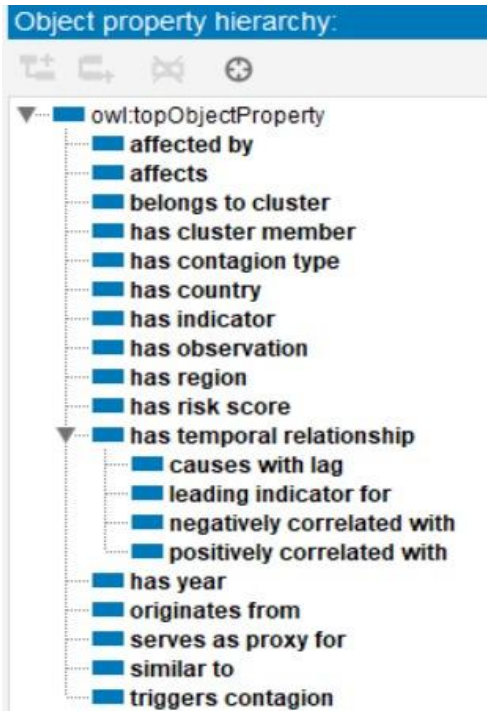
Detailed Description



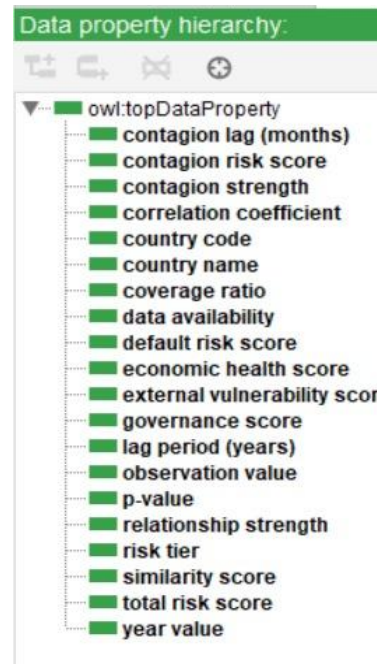
1 Classes



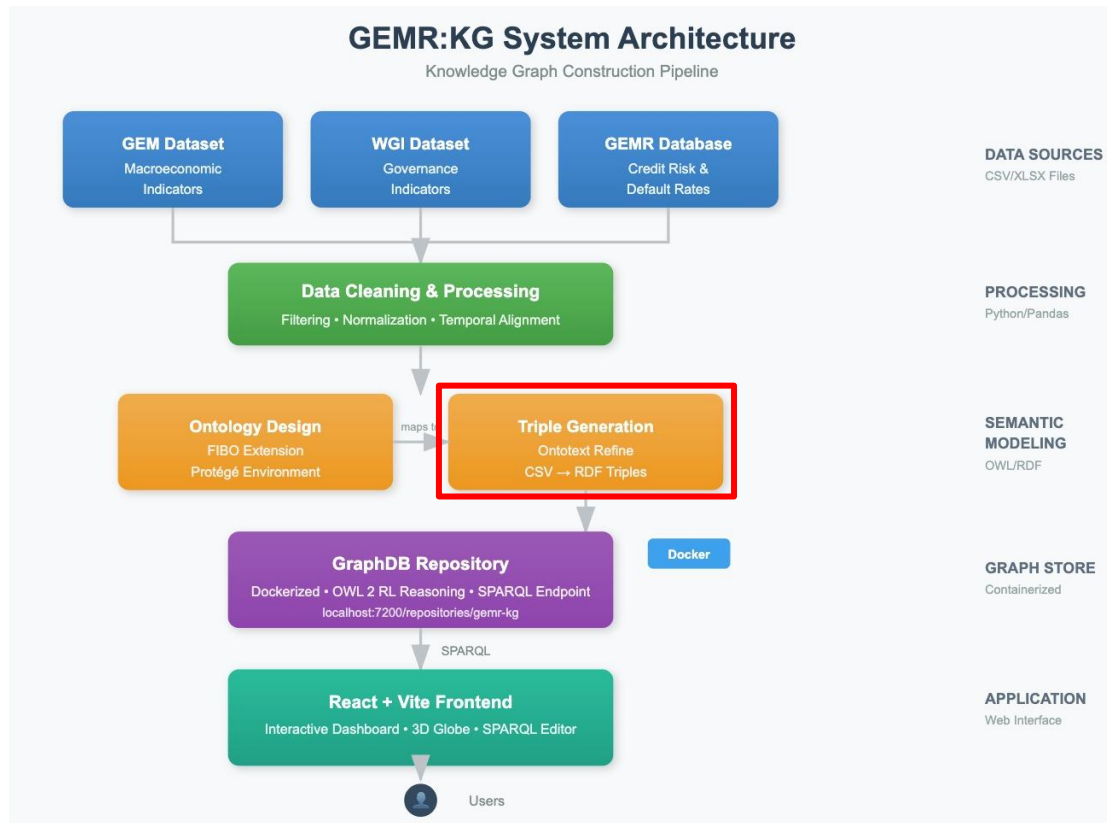
2 Object Properties



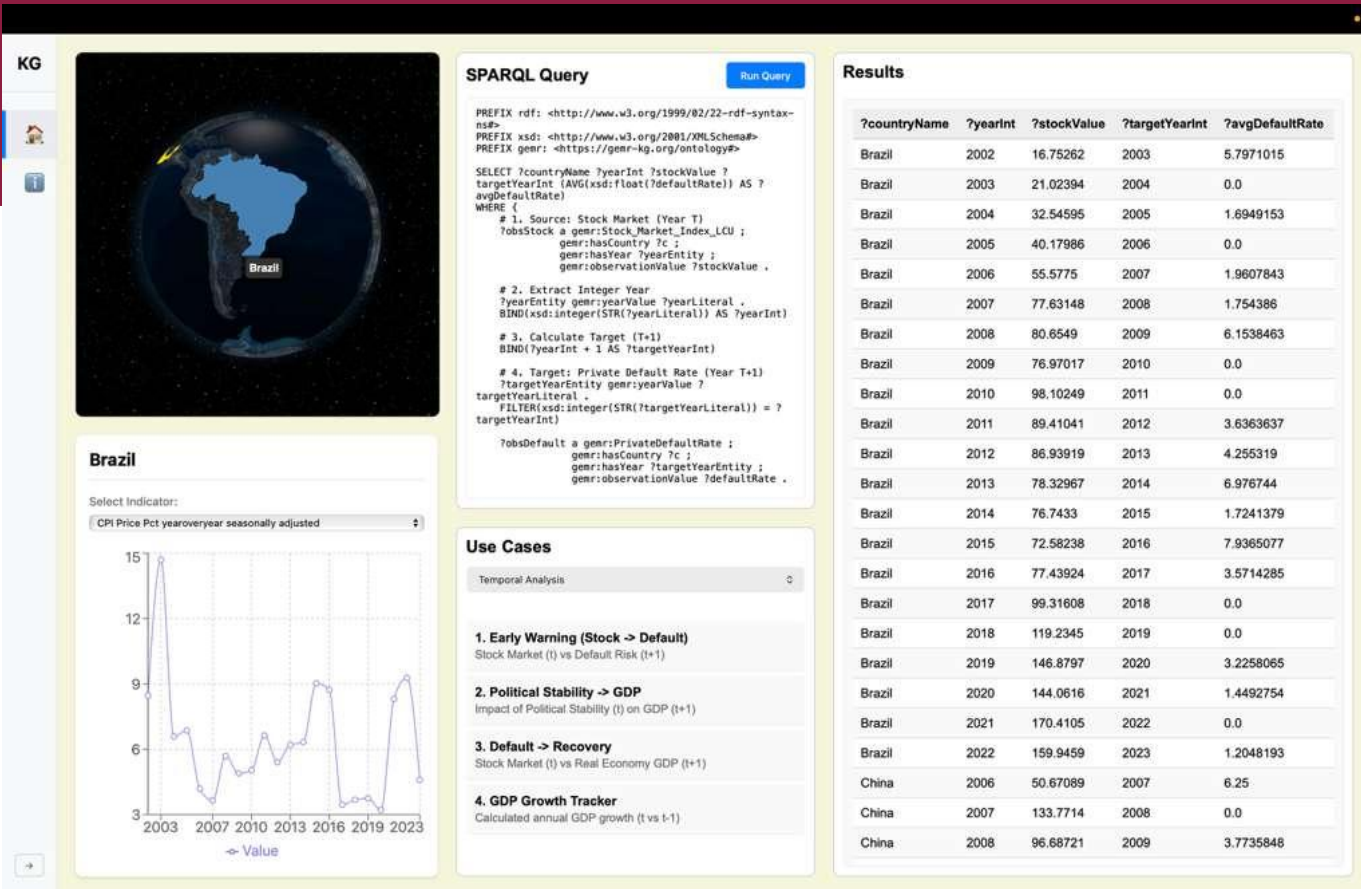
3 Data Properties



Technical Framework



User Interface



Frontend:



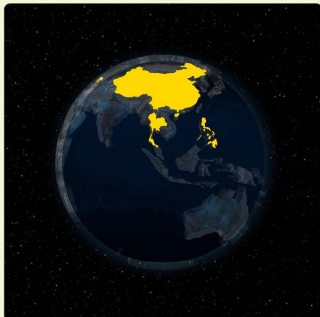
Backend:



UI Design

Detailed Description

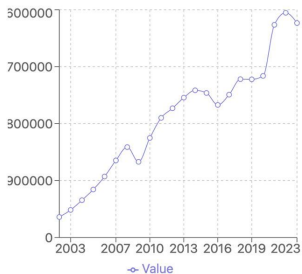
1 Indicator Tracker



China

Select Indicator:

Exports Merchandise current US seasonally adjusted



2 Use Cases

Use Cases

Contagion Analysis

Trade-Based Contagion

GDP (Source t) -> Exports
(Partner t) -> GDP (Partner t+1)

Currency Crisis Early Warning

Regional currency contagion spillover analysis

3 Query Editor

SPARQL Query

Run Query

```
PREFIX rdf:
<http://www.w3.org/1999/02/
22-rdf-syntax-ns#>
PREFIX xsd:
<http://www.w3.org/2001/XMLSchema#>
PREFIX gemr: <https://gemr-
kg.org/ontology#>
```

```
SELECT ?sourceCountry ?
targetCountry ?year
?avgSourceDepr ?
avgTargetDepr
(IF(?avgTargetDepr >
10, "HIGH (Spillover
Confirmed)",
IF(?avgTargetDepr >
5, "MODERATE (Stress)",
"LOW (Resilient)"))
AS ?
contagionRiskLevel)
WHERE {
{
SELECT ?
sourceCountry ?
targetCountry ?year
{AVG(?
sourceDepreciation_Pct) AS
?avgSourceDepr}
{AVG(?
targetDepreciation_Pct) AS
?avgTargetDepr}
```

4 Results

Results

? sourceCountry	? targetCountry	? year	? avgSourceDepr	? avgTargetDepr	? contagionRiskLevel
Brazil	Mexico	2015	41.516884	19.326132	HIGH (Spillover Confirmed)
Brazil	Mexico	2020	30.750206	11.696859	MODERATE (Stress)
Brazil	Mexico	2012	16.703415	5.735625	LOW (Resilient)

SPARQL Queries:

1. Comprehensive Data Retrieval

Fetch complete dataset (all indicators, years, values) for a specific country. Mirrors the IMF "Signal Approach" for aggregating risk factors ([International Monetary Fund 2017](#)).

```
PREFIX gmr: <https://gmr-kg.org/ontology#>
SELECT DISTINCT ?indicator ?year ?value WHERE {
  ?obs a gmr:Observation ; gmr:hasCountry ?c ;
  gmr:hasIndicator ?indicator ;
  gmr:hasYear ?yEnt ; gmr:observationValue ?val .
  ?c gmr:countryName "Brazil" . ?yEnt gmr:yearValue ?
  year .
} ORDER BY ?indicator ?year
```

2. Comprehensive Risk Profiling

Retrieves total risk score, classification, and component scores. Provides a holistic snapshot of stability vs specific areas of concern.

```
PREFIX gmr: <https://gmr-kg.org/ontology#>
SELECT ?cName ?year ?totalRisk ?gov ?econ ?vuln WHERE {
  BIND("2023"^^xsd:Year AS ?tYear) BIND("Poland" AS ?
  tCtry)
  ?sObs a gmr:RiskScore ; gmr:hasCountry ?c ;
  gmr:hasYear ?yEnt ; gmr:totalRiskScore ?total .
  ?c gmr:countryName ?cName . ?yEnt gmr:yearValue ?year
  .
  FILTER ( ?year = ?tYear && STR(?cName) = ?tCtry )
  OPTIONAL { ?sObs gmr:riskTier ?tier }
  OPTIONAL { ?sObs gmr:governanceScore ?gov }
  OPTIONAL { ?sObs gmr:economicHealthScore ?econ }
  OPTIONAL { ?sObs gmr:externalVulnerabilityScore ?vuln
  }
}
```

3. Early Warning: Stock vs Default

Analyzes Stock Market (Year T) vs Private Default Rates ($T + 1$). Tests hypothesis that market sentiment precedes fundamental credit ratings ([Wu 2008](#)).

```
SELECT ?cName ?yInt ?stock (AVG(?defRate) AS ?avgDef)
WHERE {
  ?obsS a gmr:Stock_Market_Index_LCU ; gmr:hasCountry ?c
  ;
  gmr:hasYear ?yE ; gmr:observationValue ?stock .
  ?yE gmr:yearValue ?yLit . BIND(xsd:integer(STR(?yLit))
  AS ?yInt)
  BIND(?yInt + 1 AS ?tYInt)
  ?tYE gmr:yearValue ?tYLit .
  FILTER(xsd:integer(STR(?tYLit)) = ?tYInt)
  ?obsD a gmr:PrivateDefaultRate ; gmr:hasCountry ?c ;
  gmr:hasYear ?tYE ; gmr:observationValue ?defRate
  .
  ?c gmr:countryName ?cName .
} GROUP BY ?cName ?yInt ?stock ?tYInt LIMIT 50
```

4. Political Stability Impact Tracker

Tracks negative Political Stability (T) impact on GDP ($T + 1$). Validates the 'Governance-Growth' transmission channel ([Aisen and Veiga 2011](#)).

```
SELECT ?cName ?yInt ?gdp (AVG(?polStab) AS ?avgPol) WHERE
{
  ?obsP a gmr:PoliticalStability ; gmr:hasCountry ?c ;
  gmr:hasYear ?yE ; gmr:observationValue ?polStab
  .
  FILTER(?polStab < 0)
  ?yE gmr:yearValue ?yLit . BIND(xsd:integer(STR(?yLit))
  AS ?yInt)
  BIND(?yInt + 1 AS ?tYInt)
  ?obsG a gmr:GDP_CONST_2010_USD ; gmr:hasCountry ?c ;
  gmr:hasYear ?gYHaw ; gmr:observationValue ?gdp .
  OPTIONAL { ?gYHaw gmr:yearValue ?gYVal }
  BIND(COALESCE(xsd:integer(STR(?gYVal)),?gYHaw) AS ?gYInt
  )
  FILTER(?gYInt = ?tYInt) ?c gmr:countryName ?cName .
} GROUP BY ?cName ?yInt ?tYInt ?gdp LIMIT 50
```

5. Default → Recovery Lag

Lag between stock market recovery and real economic (GDP) recovery after default. Asset prices often rebound earlier than real indicators ([Kalesnik and Polychronopoulos 2020](#)).

```
SELECT ?cName ?yT ?yTarget ?gdp (AVG(?stk) AS ?avgStk)
WHERE {
  ?obsS a gmr:Stock_Market_Index_LCU ; gmr:hasCountry ?c
  ;
  gmr:hasYear ?yT_Raw ; gmr:observationValue ?stk
  .
  ?obsG a gmr:GDP_CONST_2010_USD ; gmr:hasCountry ?c ;
  gmr:hasYear ?yT_Raw ; gmr:observationValue ?
  gdp .
  OPTIONAL { ?yT_Raw gmr:yearValue ?yVT }
  BIND(COALESCE(xsd:integer(STR(?yVT)), ?yT_Raw) AS ?yT)
  BIND(COALESCE(xsd:integer(STR(?yT_Raw)), ?yT_Raw) AS ?
  yTarget)
  FILTER(?yTarget = ?yT + 1) ?c gmr:countryName ?cName .
} GROUP BY ?cName ?yT ?yTarget ?gdp
```

6. GDP Growth Tracker

Calculates annual GDP growth percentage. Provides a standardized metric for comparing economic performance across economies.

```
SELECT ?cName ?year ?gdpCurr ?growthPct WHERE {
  ?obsG_T a gmr:GDP_CONST_2010_USD ; gmr:hasCountry ?c ;
  gmr:hasYear ?yT_R ; gmr:observationValue ?
  gdpCurr .
  ?obsG_P a gmr:GDP_CONST_2010_USD ; gmr:hasCountry ?c ;
  gmr:hasYear ?yP_R ; gmr:observationValue ?
  gdpPrev .
  OPTIONAL { ?yT_R gmr:yearValue ?yVT }
  OPTIONAL { ?yP_R gmr:yearValue ?yVP }
  BIND(COALESCE(xsd:integer(STR(?yVT)), ?yT_R) AS ?year)
  BIND(COALESCE(xsd:integer(STR(?yVP)), ?yP_R) AS ?yPrev)
  FILTER (?yPrev = ?year - 1)
  BIND(((?gdpCurr - ?gdpPrev)/?gdpPrev)*100 AS ?growthPct)
  ?c gmr:countryName ?cName .
} LIMIT 100
```

7. Trade-Based Contagion

Traces economic shocks from Source to Dependent Partner via trade. Identifies spillover risks from major trading partners ([Van Rijckeghem and Weder 2003](#)).

```
SELECT ?src ?ptrn ?yT (AVG(?sGro) AS ?sGr) (AVG(?pExGr)
AS ?pEx)
WHERE {
  { ?ptrn gmr:similarTo ?source . } UNION
  { ?source gmr:belongsToCluster ?cl . ?ptrn gmr:
  belongsToCluster ?cl .
  FILTER(?source != ?ptrn) } UNION
  { ?source gmr:countryName ?sN . ?ptrn gmr:countryName
  ?pN .
  FILTER ((STR(?sN)="Brazil" && STR(?pN)="Mexico")) }
  # Growth calc omitted for brevity
  ?source gmr:countryName ?src . ?ptrn gmr:countryName ?
  ptrn .
} GROUP BY ?src ?ptrn ?yT LIMIT 50
```

8. Currency Crisis Early Warning

Detects regional currency contagion based on "Neighborhood Effects" in emerging markets ([Basu 2001](#)).

```
SELECT ?src ?tgt ?yr ?sDep ?tDep ?risk WHERE {
  { SELECT ?src ?tgt ?yr (AVG(?sD_P) AS ?sDep) (AVG(?tD_P)
  AS ?tDep)
  WHERE {
    ?obsS a gmr:Exchange_rate_new_LCU_per_USD ;
    gmr:hasCountry ?s ; gmr:observationValue ?sV
    .
    # Calculation logic omitted
    VALUES (?sN ?tN) { ("Brazil" "Mexico") ("Thailand" "
    Philippines") }
    ?s gmr:countryName ?sN . ?t gmr:countryName ?tN .
    ?s gmr:countryName ?src . ?t gmr:countryName ?tgt
    .
  } GROUP BY ?src ?tgt ?yr }
  BIND(IF(?tDep > 10, "HIGH", IF(?tDep > 5, "MOD", "LOW")))
  AS ?risk
}
```

Sources :

Aisen, A. and Veiga, F. J. (2011). How does political instability affect economic growth? IMF Working Paper WP/11/12, International Monetary Fund.

Basu, T. G. (2001). What drives contagion? Trade, neighborhood, or financial links? IMF Working Paper WP/01/29, International Monetary Fund.

International Monetary Fund (2017). Assessing country risk: Selected approaches. Technical note, International Monetary Fund, Washington, D.C.

Kalesnik, V. and Polychronopoulos, A. (2020). Value in recessions and recoveries. Technical report, Research Affiliates.

Van Rijckeghem, C. and Weder, B. (2003). Spillovers through banking centers: A panel data analysis of bank flows. Journal of International Money and Finance, 22(4):483-509. See also IMF Working Paper 00/55.

Wu, T. (2008). The dynamic pricing of sovereign risk in emerging markets: Fundamentals and sentiment. Journal of International Money and Finance, 27(7):1171-1188.

Use Cases

Enable comprehensive time-series analysis to track lagged economic and governance events.

Use Cases

Temporal Analysis

1. Early Warning (Stock -> Default)

Stock Market (t) vs Default Risk (t+1)

2. Political Stability -> GDP

Impact of Political Stability (t) on GDP (t+1)

3. Default -> Recovery

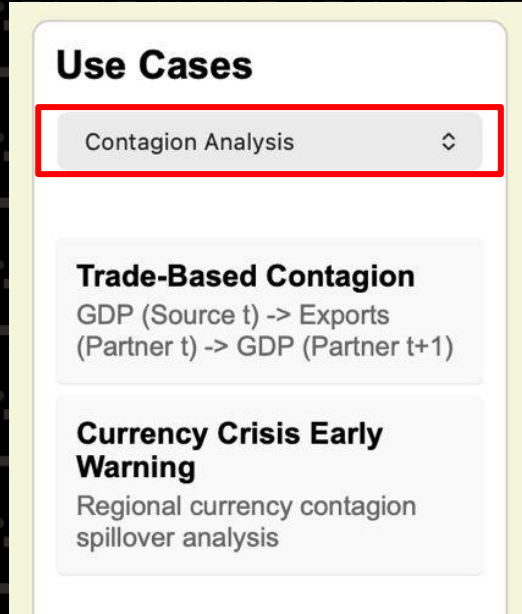
Stock Market (t) vs Real Economy GDP (t+1)

4. GDP Growth Tracker

Calculated annual GDP growth (t vs t-1)

1. Analyze time-series data to test if current market sentiment predicts future defaults.
2. Track longitudinal time-series showing political instability delayed impact on future GDP.
3. Measure time-series lags between early financial asset rebounds and actual economic recovery
4. Generate annual time-series growth metrics to standardize and compare cross-country performance

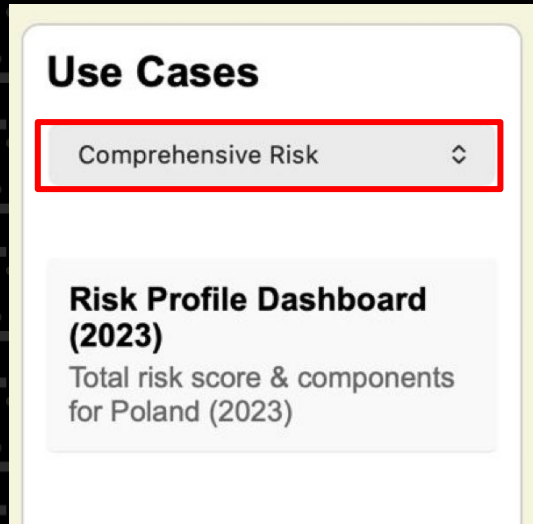
Use Cases



Model how financial distress and systemic risks spread across geographic borders.

1. Trace economic shocks transferring from a source country to a dependent trading partner.
2. Detect regional currency spillover risks based on emerging market neighborhood effects.

Use Cases



Integrate multiple economic and governance dimensions into a holistic country risk profile.

1. Retrieve total risk scores and component metrics to instantly assess a country's overall stability.

Conclusion and Strategic Significance

- Empower **researchers** and **analysts** to visualize systemic vulnerabilities and move beyond isolated statistics to assess true country-level risk.
- Integrate **isolated data** into a single, semantically **interoperable** knowledge graph.
- Leverage knowledge graphs to explicitly **model** and query **intricate** and **sophisticated** relationships.

Challenges

Pain Points within current Research



Data Scarcity

- Paywalled Data
- Public Data has continuity gaps.
- No access at all



Temporal Restriction

- Limited to 2002 - 2023
- Cannot model prior events
- Not real time.



Geographic Limitation

- Only 6 countries
- Missing Data
- Data Collection is difficult
- No standard measure/score for an indicator



Systemic Contagion

- Limited modelling support
- Difficulty tracing cross-border impacts
- Hidden risk propagation patterns

Results

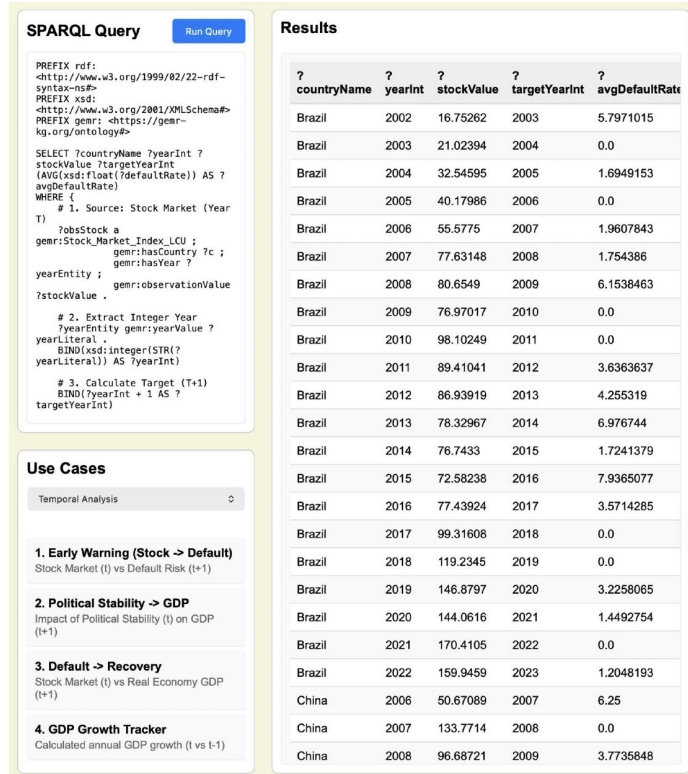


Fig. 6. Early Warning - Stock Market (t) vs Default Risk (t+1)

Results

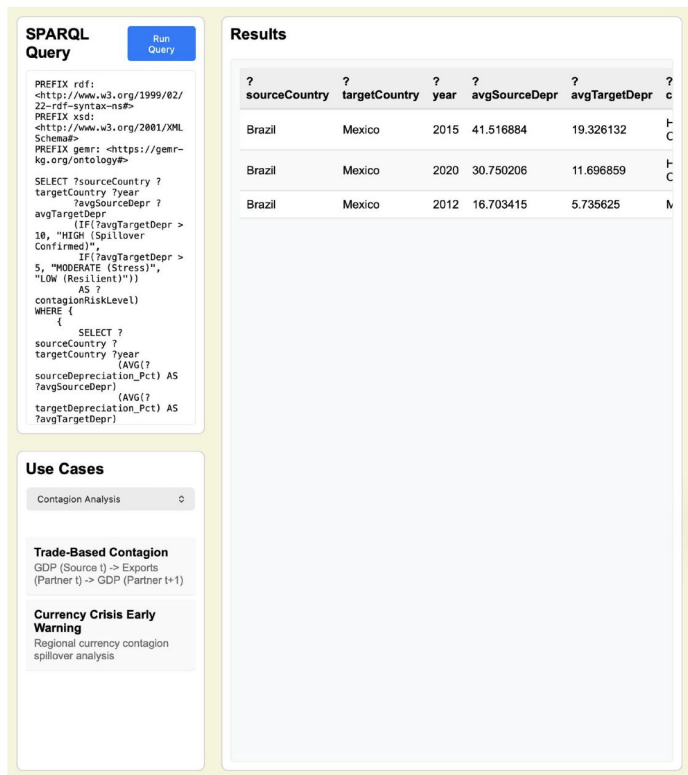


Fig. 7. Currency Crisis Early Warning - Regional currency contagion spillover analysis.

Results

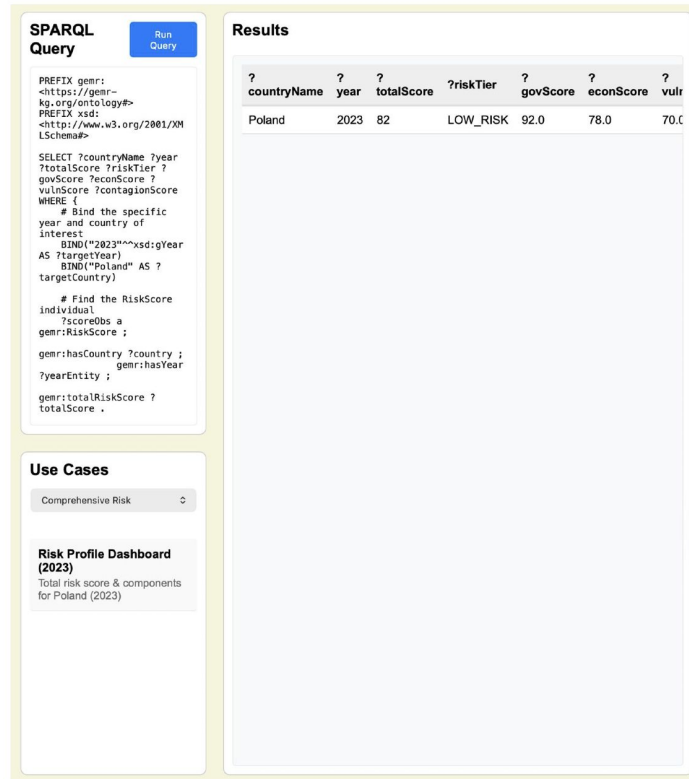


Fig. 8. Risk Profile Dashboard (2023) - Total risk score & components for Poland (2023)

Future Research Objectives

Beyond the current Scope



LLM Integration

- Natural language to SPARQL translation for non technical users



Advanced Visualizations

- Heatmaps
- Correlation Matrices



Partner with Organizations

- IFC
- World Bank
- Private Organizations
- Governments

